MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special

Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCCSC200

DATABASE MANAGEMENT SYSTEMS

(2024 ADMISSION ONWARDS)

Duration:1 Hour 30 Minutes **Maximum Marks: 50**

Part A – Very Short Answer Questions

Answer all questions
(2 × 5 = 10 Marks)

1. Associate the DBA's role with database security. (2 Marks)

Expected Answer:

- Controls user access through authentication and authorization. (1)
- Defines privileges/roles, manages backups, encryption, auditing and recovery to protect data. (1)

2. Comprehend the term Tuple of a Relation. (2 Marks)

Expected Answer:

- A tuple is a row (record) in a relation (table). (1)
- It represents one complete set of attribute values describing a single entity. (1)

3. List any four basic SQL data types and their uses. (2 Marks)

Any four:

- INT/INTEGER – stores whole numbers.
- CHAR – fixed-length characters.
- VARCHAR – variable-length strings.
- DATE – stores dates.
- FLOAT/REAL – decimal values.
- BOOLEAN – logical values.

Marking:

- 0.5 mark each (Any four × 0.5 = 2)

4. Differentiate between Stored and Derived Attribute. (2 Marks)

Stored Attribute	Derived Attribute
Physically stored in database	Calculated from other attributes
Example: Date of Birth	Example: Age

Marking

- Definition of Stored attribute – 1 , Definition of Derived attribute with example – 1

5. Define Durability. (2 Marks)

Expected Answer

- Durability is one of the ACID properties. (1)
- Once a transaction is committed, the changes remain permanent even if system failure occurs. (1)

Part B – Short Answer Questions

Answer any Five

(5 × 5 = 25 Marks)

6. Describe Transaction Management in DBMS. (5 Marks)

Scheme

- Definition of transaction management – 1
- Ensures ACID properties – 2
- Controls concurrent transactions – 1
- Recovery from failures (commit/rollback) – 1

7. Discuss Armstrong's Axioms with examples. (5 Marks)

Scheme

- Reflexivity – 1
- Augmentation – 1
- Transitivity – 1
- Suitable examples – 2

8. Illustrate INSERT INTO...SELECT command. (5 Marks)

Scheme

- Purpose of INSERT INTO...SELECT – 2
- Correct SQL syntax – 1
- Suitable example – 2

9. Describe Entities and Attributes. (5 Marks)

Scheme

Definition of Entity – An entity in DBMS represents a real-world object or concept that can be uniquely identified and stored in a database. It forms the foundation of data modeling in an Entity-Relationship (ER) model.(1)

Example: A student, An employee, or bank a/c, etc. all are entities.(1)

Types of Entities such as strong,weak,Derived,Multivalued etc.(1)

Definition of Attribute – An Attribute is a property or characteristic that describes an entity in a database. It provides specific information used to identify, categorize, and manage entities effectively. Without attributes, entities have no meaningful data.(1)

Example –Single, Composite ,Multivalued etc. (1)

10. Explain Integrity Constraints. (5 Marks)

Expected Types

- Domain Constraint
- Entity Integrity
- Referential Integrity
- Key Constraint

Scheme

- Any four constraints (1 mark each) – 4
- Suitable examples – 1

11. Explain INNER JOIN and write SQL query. (5 Marks)

Scheme

- **INNER JOIN** is used to combine rows from two or more tables based on a related column. It returns **only the rows where there is a matching value in both tables**. (1)
- Demonstrate Employee and Department Tables with a common attribute.(2)
- Correct SQL syntax of INNER JOIN – (2)

12. Difference between UNION and UNION ALL. (5 Marks)

UNION	UNION ALL
Removes duplicates	Keeps duplicates
Slower	Faster
Distinct result	All records returned

Scheme

- Definition of UNION – 1
- Definition of UNION ALL – 1
- Comparison – 3

Part C – Essay Question

Answer any One

(15 × 1 = 15 Marks)

13. Draw an ER Diagram of a Company Database. (15 Marks)

Scheme

- Identification of entities (Employee, Department, Project etc.) – 3
- Attributes – 3
- Relationships – 3
- Correct cardinalities – 3
- Neat ER diagram with proper notation – 3

14. Explain Normal Forms and Normalization. (15 Marks)

Scheme

- Definition of normalization – 2
- Primary key concept – 2
- First Normal Form (1NF) Rules and Process – 3
- Second Normal Form (2NF) Rules and Process – 3
- Third Normal Form (3NF) Rules and Process – 3
- Appropriate examples – 2